Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Next-generation process design for a cost-efficient glass substrate and system production for 3D Augmented Reality Head-Up-Displays*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Machinery | materials		*Machinery | materials*	
	Medium Category	Small Category	Medium Category	Small Category
	Precision manufacturing machinery	*Press machines*	*Automobile*	*Vehicle Intelligence Technologies*
Concept of Proposed Technology	*The core concept is a cost-efficient process for producing high-precision glass substrates for 3D AR Head-Up-Displays, using advanced isothermal glass molding techniques. This approach enables the replication of complex glass geometries with excellent form accuracy and surface quality, eliminating the need for post-processing such as polishing.*			
Rationale for Joint R&D Project	*This joint R&D project is needed to address the technical challenges of scalable, high-quality glass forming and coating for advanced AR HUD applications. Collaboration is essential to combine Fraunhofer IPT's expertise in process development with Korean SMEs' capabilities and market access. Together, the partnership accelerates innovation, ensures technology transfer, and enables rapid commercialization for global competitiveness.*			
Key Words	1. Precision Glass Molding 2. Thin Glass Forming 3. Head-Up-Display 4. Cost-effective production			

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 5	End of the Project	Level 7

R&D Objectives

Glass is a unique material that exhibits several advantages over plastics, making it particularly suitable for display panels. It is scratch-resistant, which increases its lifespan, and offers good tactile properties. Glass withstands environmental influences such as UV radiation and acids over the long term. Additionally, it can be recycled without a loss of quality, whereas plastics often have limited recycling potential. The high transparency of glass ensures excellent light transmission, which enables it to be used for a wide range of applications as shown in figure 1. [STRO22; VU16]

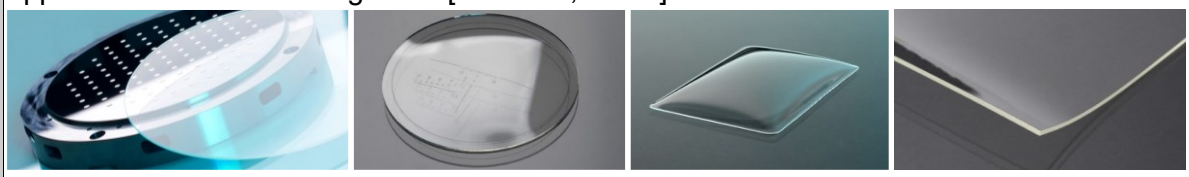

Figure 1: Wafer-level optics, microstructured glass, cover glass, center console

Glass is extensively applied (compare figure 2) in different technologies across various sectors. In automotive interiors, glass is utilized in components such as center consoles, speaker cover, mirror replacement systems, or **head-up-display mirror substrates**. In the realm of consumer electronics, glass is commonly employed in devices like smartphones and smartwatches, where its scratch resistance and high transparency contribute to the functionality and visual clarity of screens. Furthermore, in the field of virtual reality (VR), glass is essential for creating immersive displays that offer high resolution and clarity. Glass can also be used in the semiconductor industry for the encapsulation of products, such as MEMS devices. Overall, the use of glass in these applications highlights its superior properties, which enhance both performance and user interaction across diverse industries.

*Figure 2: Applications for thin glass (left to right: speaker cover, semiconductor MEMS cover, AR/VR, **Head-Up-Display,** automotive interior) [BANG24; FLAB23; FRAU24; GUAY23]*

Multiple market forecasts indicate a strong anticipated demand for 3D AR HUD, projecting a compound annual growth rate (CAGR) of approximately 15.6 % as seen in figure 3 [MARK24]. This trend reflects not only the growing popularity of HUDs but also to other industries. Potential applications extend to sectors such as displays, sensor elements, and semiconductors, highlighting the expanding market opportunities. As the pressure to reduce emissions intensifies, energy-efficient technologies, such as the proposed thin glass forming process, emerge as a core driver in achieving low-emission production. By optimizing use of resources and minimizing waste, these technologies not only contribute to sustainability goals but also align with the Paris Agreement to limit worldwide temperature increase to 1.5 °C [CLIM24].

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Figure 3: Market potential of HUD devices (left), Display application (centre), Head-up-display application (right) [CLIM24; CONT25; FINK22; MARK24; RESE22]

The current state of the art for forming of thin glass (via the press bending process) involves a process chain characterized by several key methods and challenges. In isothermal forming, both the glass and the tooling are maintained at approximately the same temperature throughout the process (compare figure 4). Additionally, the use of graphite molds in conjunction with long contact times between the glass and the mold, necessitates subsequent polishing to achieve the desired surface quality since this material leaves marks and air bubbles on the glass. This leads to high scrap rates, as the glass often breaks during the polishing process, resulting in significant waste.

The proposed process chain introduces innovative approaches. By using steel or other mold materials, the cost efficiency dramatically increases compared to conventional materials such as graphite. This new approach eliminates the need for polishing, as the application of alternative tool materials allows for direct use of the finished component. Additionally, vacuum-assisted slumping techniques can be applied to introduce a force (negative pressure) alongside the gravitational force, improving cycle time.

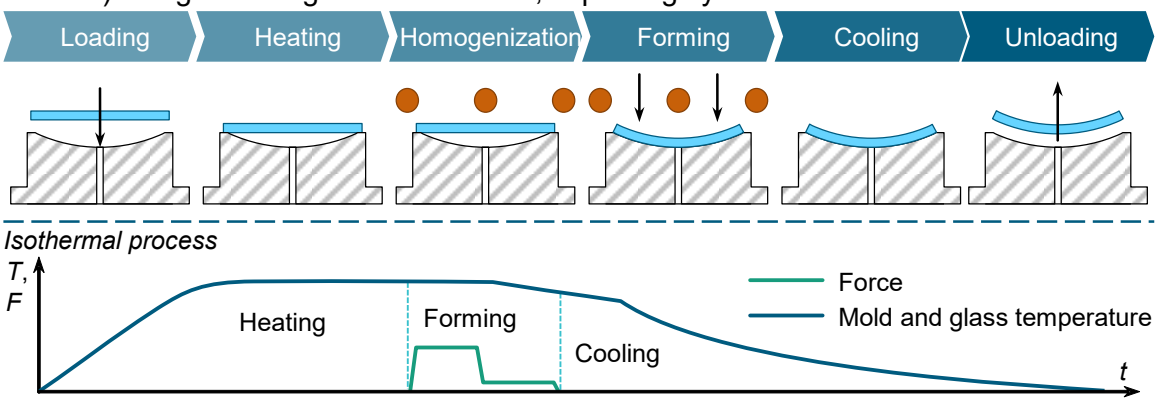

Figure 4: Precision Glass Molding for Thin Glass Substrates

To achieve a competitive advantage in glass forming technology, it is crucial to implement cost-efficient production methods, especially for Korean SMEs. In an ever-increasing globalized world, it is crucial to achieve a certain degree of technological and political independence in order to ensure resilience against external crises. Furthermore, there is significant innovation potential to drastically transform the production processes of thin glass products. As competition intensifies, industries must not only embrace advanced technologies and innovative approaches but also remain vigilant of global market trends and competitor strategies. By enhancing efficiency and reducing costs companies can ensure long-term viability and competitiveness. This proposed process chain holds high potential for energy- and cost-efficient production.

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R&D Target Specifications	**Technical goals** - Enabling sustainable production by digitalization techniques - Degree of automation: high - Form accuracy: < 30 µm (200 mm x 100 mm) - Surface roughness: < 10 nm - Estimated cost per unit: < 5,00 EUR (~8000 KRW) at 100,000 units per year - Emissions per unit: < 3 kg_{CO2} - Cycle time: < 5 min - No polishing after forming, thus reducing post-processing and increase of yield (no air bubbles, no marks) - Transfer of knowledge and licensing to Korean SME - Deployment of technology to other geometries and applications
R&D Plan	*Work Package (WP) 0: Project management (Korean partners)* Project management will oversee the entire project duration, ensuring that all work packages are executed efficiently and within the established timeline. **Phase I:** *WP 1: Definition of demonstrator and requirement analysis* (Korean partners) Focus will be on defining the demonstrator, including the geometry of the glass, HUD design, and intended application. Comprehensive documentation will be created to outline the specifications and objectives for each demonstrator. *WP 2: Pre-study and preparation of phase II (All partners)* Pre-study and preparation of phase II involves benchmarking current materials and processes relevant to thin glass forming and AR HUD applications. Initial feasibility studies and risk assessments will be conducted, including preliminary process simulations and small-scale experiments to validate key concepts. The outcome will be a readiness report and a detailed plan for the main development phase. **Phase II:** *WP 3: Optical simulation and system design (Korean partners)* Optical simulation and system design focuses on developing and optimizing the optical layout for 3D AR Head-Up-Displays. Advanced optical simulations will be performed to evaluate system performance, assess substrate geometry, and ensure compatibility with production requirements. The results will guide the design of demonstrator and integration of glass substrates into AR HUD systems. *WP 4: Glass Forming (Fraunhofer IPT)* This work package will focus on the manufacturing processes specific to glass substrates, encompassing various aspects of production. *WP 4.1: Simulation of molding process* The molding process will be simulated by using FEM modeling technique to predict the behavior of glass during the forming process and to derive the molding tool design. This allows the identification of potential challenges and optimization of parameters before physical production begins. *WP 4.2: Design and manufacturing of molding tools* Based on the simulation in WP4.1 this sub-work package will focus on the design and

fabrication of specific molding tools required for the innovative glass forming processes.

WP 4.3: Process development of thin glass forming process

This sub-work package will concentrate on developing and refining the thin glass forming process, focusing on achieving the desired geometries and quality standards. Experimental trials will be conducted to investigate various parameters.

WP 4.4: Qualification

Metrology will play a critical role in ensuring the quality and accuracy of the produced glass displays. This sub-work package will establish measurement protocols and techniques to monitor key parameters throughout the manufacturing process.

WP 5: Development of glass coating (Korean partners)

Development of glass coating will focus on designing and applying functional coatings, including reflective aluminum coatings, to the glass substrates. Coating processes will be optimized to achieve high reflectivity, durability, and optical quality required for AR HUD applications. The effectiveness of the coatings will be validated through performance testing and integration trials.

WP 6: System integration and evaluation (Korean partners)

System Integration and Evaluation will involve assembling the coated glass substrates into the complete AR HUD system and ensuring compatibility with existing technologies. Functional and performance tests will be conducted to validate optical quality, durability, and system reliability. Feedback from these evaluations will guide final adjustments and confirm readiness for mass production.

The Gantt-chart, as seen in figure 5, will facilitate effective project management.

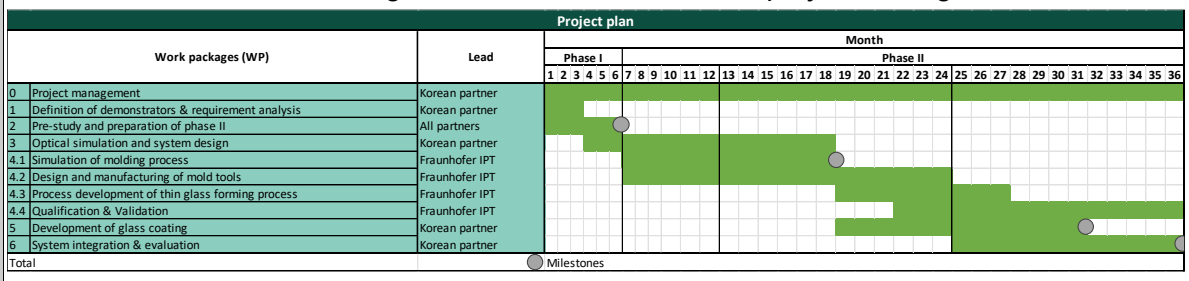

Figure 5: Proposed Gantt-chart of the project

Milestone 1: Finalize a comprehensive list of requirements and an evaluation catalog, ensuring stakeholder agreement on all criteria. Assess and document the results of Phase I to enable detailed planning and targeted execution of development tasks in Phase II.

Milestone 2: Complete the development of a simulation model that accurately predicts the glass forming process, validated through at least three test scenarios.

Milestone 3: Coating developed and validated on glass substrates, meeting optical and durability requirements for AR HUD applications.

Milestone 4: Successful system integration of coated glass substrates into AR HUD demonstrators, with functional and performance validation completed.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**Fraunhofer IPT** will develop the forming technology regarding thin glass forming, e.g. for glass head-up-display mirror substrates. IPT utilizes simulations to optimize the forming process, designs and manufactures the necessary forming tools and holds the capability to develop the production process in-house. Contributions from Fraunhofer IPT follow in detail: <u>Simulation / Digital twin</u> The project emphasizes the simulation of the molding process through finite element modeling (FEM) techniques, which predict the behavior of glass during forming and serves as a digital twin of the process. This approach allows for the identification of potential challenges and the optimization of various parameters prior to the commencement of physical production, e.g. mold compensation by simulation. By simulating the forming process, the project aims to enhance efficiency and reduce risks associated with the manufacturing of glass displays. <u>Process development and metrology</u> The development and refinement of the thin glass forming process are central to achieving the desired geometries and quality standards. This involves conducting experiments to test and optimize various parameters, ensuring that the production process aligns with high-quality benchmarks, such as air bubble free surfaces. Additionally, the establishment of measurement protocols will play a critical role in maintaining precision and quality throughout the manufacturing process, contributing to the overall success of the project. <u>Transfer of knowledge to Korean industry</u> A crucial aspect of the project is the transfer of knowledge among stakeholders, particularly through collaboration between experts from Fraunhofer IPT and industry partners. This exchange of expertise will facilitate the sharing of best practices, innovative techniques, and lessons learned from the simulation and process development phases. By fostering a collaborative environment, the project aims to enhance the skill sets of participants and promote the adoption of advanced manufacturing processes in the glass display industry. This knowledge transfer is essential for driving innovation, improving production efficiency, and ensuring the long-term sustainability of the technology developed within the project.
Expected R&R of Korean SME Partner	**Korean partners** are responsible for integrating technology into the prototype production of curved glass substrates, including the definition of the requirements. They will adapt manufacturing processes, collaborate on prototype refinement, and conduct tests to ensure the designs meet performance and aesthetic standards, facilitating the transition to mass production. Contributions from Korean partners in detail: <u>Project management</u> Project management will oversee the entire duration of the project, ensuring that all work packages are executed efficiently and within the established timeline. This structured approach is essential for coordinating activities, managing resources, and mitigating risks, thereby facilitating the successful completion of project objectives. <u>Definition of demonstrator & requirement analysis</u> The Korean partner will emphasize the definition of demonstrator, focusing on the geometry

of the glass substrate, display design, or other intended applications. Comprehensive documentation will outline the specifications and objectives for each demonstrator, while a thorough requirement analysis will examine both technical and user needs. This process will involve creating detailed requirements and functional specifications, with stakeholder input solicited to ensure alignment with industry standards and user expectations.

Optical simulation

The Korean partners will be responsible for conducting the optical simulation work. This includes designing and optimizing the optical layout for the 3D AR Head-Up-Displays, performing advanced simulations to assess system performance, and ensuring that the substrate geometry and coatings meet the required optical specifications. Their expertise in optical modeling will be crucial for validating the demonstrator designs and supporting the successful integration of the glass substrates into the final application.

Optical coating development

The development of glass coatings focuses on designing and applying functional layers, such as highly reflective aluminum coatings, directly onto the formed glass substrates. The coating process will be optimized for high reflectivity, durability, and optical quality, meeting the specific requirements of 3D AR Head-Up-Display applications. Performance testing and integration trials will validate the coatings to ensure they deliver reliable results in demanding automotive environments.

Integration into current manufacturing technologies

One of the final phases of the project will concentrate on integrating the developed glass manufacturing processes into existing manufacturing technologies. Collaboration with technology providers will be essential to ensure compatibility and enhance overall production efficiency. Rigorous testing and validation will confirm that the integrated processes meet established performance and quality standards, paving the way for successful commercialization.

Ramp-up to mass production

Following the integration phase, the project will facilitate a ramp-up to mass production, addressing scalability and operational challenges. This stage will involve refining processes based on initial production feedback, optimizing workflows, and ensuring that the manufacturing infrastructure is capable of meeting market demands. The transition to mass production will be critical for realizing the commercial potential of the developed technologies and achieving widespread adoption in the industry.

System integration

The Korean partners will also be responsible for system integration. This involves assembling the coated glass substrates into the complete AR Head-Up-Display system, ensuring compatibility with existing technologies, and conducting functional and performance testing. Their role in system integration is essential to validate the overall performance, reliability, and readiness of the technology for mass production and commercialization.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	The technical targets for curved glass forming development focus on achieving high precision and excellent surface quality to ensure sufficient form accuracy in the final product. Additionally, cost efficiency is essential to reduce production expenses while maintaining quality, and energy efficiency aims to minimize energy demand during the forming process. High precision and surface quality are essential, and the use of steel molds eliminates the need for polishing. **Enabling efficient thin glass forming for Korean industry partners** This project emphasizes an isothermal forming process that optimizes cycle times and improves product quality, eliminating the need for post-processing. By leveraging replicative isothermal thin glass forming, the project aims to achieve high precision and surface quality while minimizing energy demand and waste. Collaboration between Fraunhofer IPT and Korean industry partners will facilitate knowledge transfer and integration of these advanced technologies into prototype production. Finally, this will empower Korean partners to adopt scalable thin glass forming technologies, enhancing their competitiveness in the global market and supporting broader environmental objectives.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization